



FHSS-Net: A novel instance segmentation network for multi-class fetal heart structures via global context and multi-scale learning

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ABSTRACT

Accurate segmentation of fetal heart ultrasound images is crucial for early screening and prenatal diagnosis of congenital heart disease (CHD). However, high noise levels, blurred anatomical boundaries, and scale variations in ultrasound images make precise localization and segmentation of key anatomical structures highly challenging. Despite significant progress in deep learning for medical image analysis, the scarcity of publicly available, high-quality fetal heart ultrasound datasets continues to limit advancement in this field. To address this challenge, we constructed FH-3 K, a specialized fetal heart ultrasound dataset covering five standard fetal heart views and providing detailed annotations for ten key anatomical structures. Using FH-3 K, we established benchmarks for multiple instance segmentation models to guide future research. Yet, attaining higher accuracy remains difficult due to the dataset's limited scale and the inherent complexities of ultrasound imaging. We thus propose FHSS-Net, a new instance segmentation network optimized for fetal heart ultrasound. FHSS-Net integrates a lightweight Vision Transformer module (EfficientViT) to enhance long-range dependency modeling and improve boundary delineation. Additionally, dilated convolutions expand the receptive field without increasing computational cost, and an E-Focal Loss mitigates class imbalance, boosting fine-grained segmentation accuracy. Under five-fold cross-validation, FHSS-Net achieves 93.7 % localization accuracy and 92 % segmentation accuracy across ten anatomical structures, outperforming mainstream models. The construction of FH-3 K and the introduction of FHSS-Net offer valuable support for fetal heart ultrasound segmentation, potentially improving CHD screening and prenatal diagnostic workflows.

1. Introduction

Congenital Heart Disease (CHD) remains a leading cause of morbidity and mortality in newborns, with a prevalence of approximately 0.8 %–1.2 %, according to the World Health Organization (WHO) and related studies[1]. Due to its complexity and diversity, early detection and accurate prenatal diagnosis of CHD are crucial, as they

enable timely interventions that significantly reduce neonatal mortality and disability[2]. These measures also help alleviate financial and emotional burdens on families and society. Ultrasound imaging is the primary method for routine prenatal examinations owing to its non-invasive nature, real-time capability, convenience, and relatively low cost[3]. Despite offering detailed visualization of the fetal heart, ultrasound examinations are frequently hindered by fetal position changes,

Abbreviations: 3V, Three vessel view; 3VT, Three vessel tracheal view; 4CH, Apical four chamber view; RVOT, Right ventricular outflow tract view; LVOT, Left ventricular outflow tract view; LA, Left atrium; LV, Left ventricle; RV, Right ventricle; RA, Right atrium; AA, Ascending aorta; SVC, Superior vena cava; AO, Aorta; PA, Pulmonary artery; MPA, Main pulmonary artery; T, Trachea.

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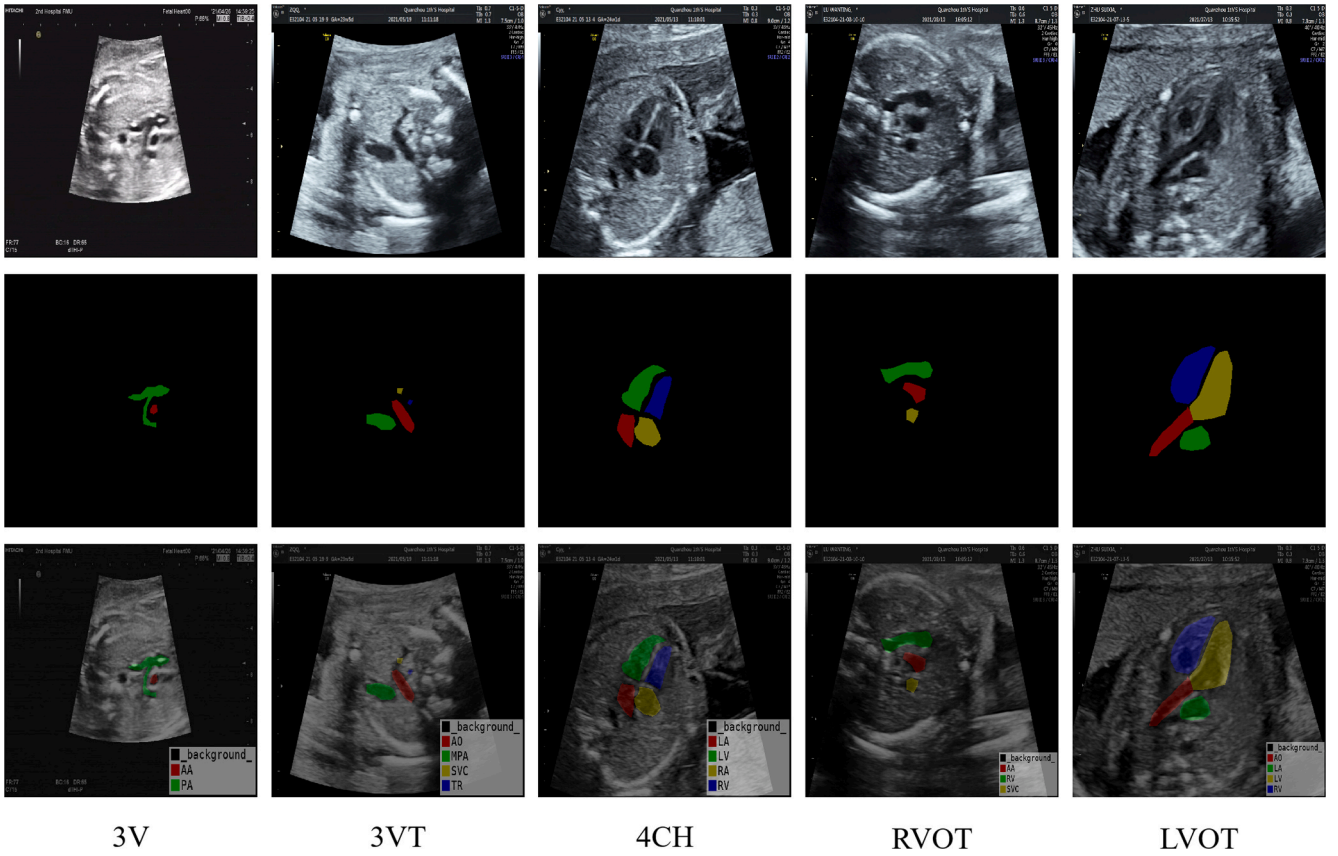


Fig. 1. Five fetal heart ultrasound standard planes, ten fetal heart key anatomical structure datasets, and annotation examples.

probe angle limitations, amniotic fluid volume variations, and maternal characteristics. These factors often lead to low signal-to-noise ratios, artifacts, uneven contrast, and blurred boundaries. Moreover, the fetal heart's small size, continuous motion, and subtle differences among its chambers, vessels, and valves make accurate recognition and segmentation challenging. Manual annotation introduces subjectivity and complicates large-scale quantitative analyses[4].

Recent advances in deep learning have driven significant progress in image segmentation and object detection[5–7]. However, semantic segmentation merely labels homogeneous regions without differentiating multiple instances of the same class, while object detection provides bounding boxes that lack the precise delineation necessary for clinical applications. In contrast, instance segmentation provides pixel-level boundaries for each anatomical structure, enabling more accurate identification of fetal heart chambers, vessels, and surrounding tissues[8]. Despite this promise, several challenges persist. In particular, the effectiveness of network feature extraction and fusion, multi-target precise localization and discrimination, as well as specialized optimization strategies for small targets and class imbalance remain pivotal technical hurdles. Additionally, the scarcity of large-scale, high-quality public ultrasound datasets—stemming from ethical and practical constraints—limits the generalizability of these models[9].

Building upon the aforementioned challenges, this paper introduces a novel network architecture that simultaneously prioritizes efficiency and high accuracy to achieve precise instance segmentation for essential fetal heart structures. To this end, we design a deep learning model named FHSS-Net. First, a lightweight yet high-performance Vision Transformer (EfficientViT) is embedded into the backbone network, leveraging the self-attention mechanism to effectively capture long-range dependencies while employing parameter reduction strategies that preserve robust global context modeling without incurring significant computational overhead. Second, dilated convolutions are adopted

in the feature fusion stage to substantially expand the receptive field without noticeably increasing the parameter count, thereby providing more comprehensive multi-scale feature support and exhibiting heightened robustness in handling ambiguous boundaries or significant variations in anatomical structures. Finally, E-Focal Loss is incorporated to enhance the model's capacity to distinguish between complex and straightforward samples and maintain focus on small targets, ensuring stable training and inference performance even under conditions of pronounced class imbalance and substantial noise.

The main contributions of this work are summarized as follows:

- (1) We collected a dataset of fetal heart ultrasound images – FH-3 K, which contains 3360 images from 5 standard fetal heart planes annotated with 10 basic anatomical structures. To our knowledge, this dataset contains the richest standard planes and structures for fetal heart ultrasound. This provides a solid foundation for deep learning in medical research. Fig. 1 shows some examples.
- (2) We introduce a novel high-precision framework that optimizes feature learning, global context modeling, and imbalanced sample handling. By integrating a lightweight EfficientViT, incorporating dilated convolutions for broader receptive fields, and employing E-Focal Loss, FHSS-Net notably boosts fetal heart structure detection and segmentation performance.
- (3) Comparative experiments and ablation studies confirm FHSS-Net's superiority over existing methods, especially in high-noise, blurred-boundary, and small-target scenarios, underscoring its potential for clinical fetal heart ultrasound applications.

The rest of the paper is organized as follows: Section 2 reviews related research. Section 3 presents the datasets and model details of the proposed FHSS-NET. Section 4 describes the detailed experimental setup

and validates its effectiveness. Section 5 summarizes the overall work and discusses possible future developments.

2. Related work

2.1. Traditional methods and early automation frameworks

Before the rise of deep learning, researchers often used traditional image processing techniques and more basic machine learning methods to deal with noise and complex tissue structures in ultrasound images.

(Anquez, 2013) integrates different distribution models to achieve segmentation of the uterus fetus unit in 3D ultrasound, utilizing regional consistency to enhance the stability of the results[10]. (Rahmatullah, 2014) uses multi-layer superpixel classification combined with the random forest method on ultrasound images to improve fetal organ boundaries' recognition ability [11]. (Vargas-Quintero, 2016) segmented the left ventricle in fetal echocardiograms using a multi-textured active appearance model based on the guided Hermite transformation[12]. (Wu, 2017) proposed a cascade framework for fully automated prenatal ultrasound image segmentation to improve the efficiency and objectivity of biometric interpretation[13].

In this stage of work, the main focus is on using prior knowledge, statistical modeling, and multi-stage strategies to resist noise interference and boundary blurring. However, when dealing with situations where there are significant differences in scale or organ morphology, problems such as insufficient adaptability and limited generalization ability remain.

2.2. Application of deep learning in specific organ and structure segmentation

With the rise of deep neural networks, researchers have begun to widely apply convolutional neural networks and their variants to automated segmentation and measurement of fetal organs, focusing on precisely identifying single organs or specific anatomical structures.

(Sundaresan, 2017) used a Fully Convolutional Network (FCN) to extract features from the fetal heart automatically and used target localization as the basis for subsequent echocardiography analysis [14]. This work emphasizes locating and refining the heart region in complex images quickly. (Yang, 2018) constructed a fully automated multi-structure segmentation framework that combines dense convolutional networks with layered deep supervision to uniformly segment multiple targets, including fetuses, amniotic sacs, and placenta, demonstrating the possibility of parallel segmentation of numerous structures[15]. (Meenakshi, 2019) developed a hybrid algorithm for segmenting images of fetal kidneys in the second and third trimesters with 86 % accuracy compared to other techniques[16]. (Xu, 2020) introduced a cascade U-net with Structural Similarity Index Measurement (SSIM) loss for automated segmentation of heart structures in ultrasound A4C views. The method aims to segment structures such as epicardium, left ventricle, left atrium, descending aorta, right atrium, right ventricle, and chest in a single pass[17]. (Qiao, 2020) achieved segmentation and HC measurement of the fetal skull using the expandable processing of U-Net architecture [18]; (Sobhaninia, 2020) and (Li, 2020) both conducted deep learning optimization on the segmentation and measurement of head-related parameters (HC, BPD, OFD)[19,20]. (Oguz, 2020) proposed a semi-automatic segmentation scheme for placental tissue[21].

Such research typically relies on the powerful advantages of convolutional neural networks in image feature extraction, focusing on using network structure refinement or loss function improvement to enhance segmentation accuracy and stability. However, due to most methods targeting individual organs or specific structures, the comprehensive problems of high noise, class imbalance, and scale differences faced in the simultaneous segmentation of multiple structures have not been fully solved.

2.3. The latest progress in multi-structure and advanced frameworks

With the rise of attention mechanisms, multi-task learning, and instance segmentation strategies, recent research has focused on processing multiple organs or key anatomical regions simultaneously in a single model and proposed higher-level technical solutions for the complexity of real clinical scenarios.

(Xi, 2021) proposed a multi-scale model combined with an attention mechanism to segment the fetal heart and lungs simultaneously[22]; (Wong, 2021) introduced spatial dropout, group normalization, and deep supervision based on V-Net to achieve multi-view and multi-category fetal ultrasound image segmentation[23]. (Budd, 2021) proposed an interpretable learning mapping method for hypoplastic left heart syndrome (HLHS) detection[24]; (Liu, 2021) focuses on the segmentation of hip joints in newborns and establishes standard plane detection[25]; (Zeng, 2021) and (Singh, 2021) implanted deep supervision mechanisms in attention V-Net and ResU-Net-c, respectively, to achieve more accurate localization in brain and cardiopulmonary segmentation[26,27]. (Hesse, 2021) conducted research on subcortical segmentation of fetal brain structure under 3D ultrasound[28]; (Nageswari, 2022) focuses on TS-GWA as the core, emphasizing heart chamber segmentation and high adaptability[29]; (Pu, 2022) Key heart structure segmentation was achieved by integrating MobileNet, UNet, and feature pyramids[30]. (Belciug, 2024) proposed an intelligent decision support system that combines deep learning and clustering techniques for fetal morphology scanning[31]; (Lu, 2024) designed the IS-YOLOX profound instance segmentation network based on YOLOX for identifying and segmenting fetal heart anatomical structures[32]; (Sengupta, 2024) explored the issue of over-segmentation in medical image segmentation using SAM series models[33]; (Li, 2024) proposed a method for detecting key fetal anatomical structures using topological knowledge reasoning to help accurately recognize ultrasound standard sections[34]. SPReCHD [35] proposed a semantic parsing network tailored for the four-chamber (4CH) view to assist in fetal CHD diagnosis, highlighting the potential of deep learning in standard-view semantic segmentation tasks. However, its focus remains limited to a single anatomical plane and four cardiac chambers, while our work targets multi-view and multi-structure instance segmentation across ten fetal heart components.

This stage of research focuses more on the practical applications of "multi-structure simultaneous segmentation," "attention and depth feature fusion," and "instance segmentation or object detection" in ultrasound images. By introducing more advanced network structures, attention mechanisms, and more targeted loss functions or optimization strategies, many works have achieved remarkable results in practical clinical scenarios with high noise, large-scale differences, and severe class imbalance, providing new technical support and development direction for the automated diagnosis and analysis of fetal ultrasound images.

Such research often follows the paradigm of relying on powerful convolutional feature extraction capabilities and continuously optimizing network structures or loss functions, in order to achieve higher accuracy and stability in multi structure segmentation and advanced network frameworks. However, the network complexity and training difficulty associated with multi-scale feature fusion and deep supervision mechanisms have also increased, limiting inference speed and making it difficult to simultaneously meet real-time and clinical deployment requirements. At the same time, there is still a lack of large-scale, high-quality, cross device, and cross center unified datasets, which poses challenges to the implementation of multi task learning and cross domain adaptation, and restricts the generalization performance of algorithms.

To address these difficulties and further promote the development of this field, this article proposes more targeted and innovative technical solutions based on existing research results to achieve breakthroughs in multi-structure segmentation accuracy, model robustness, and clinical

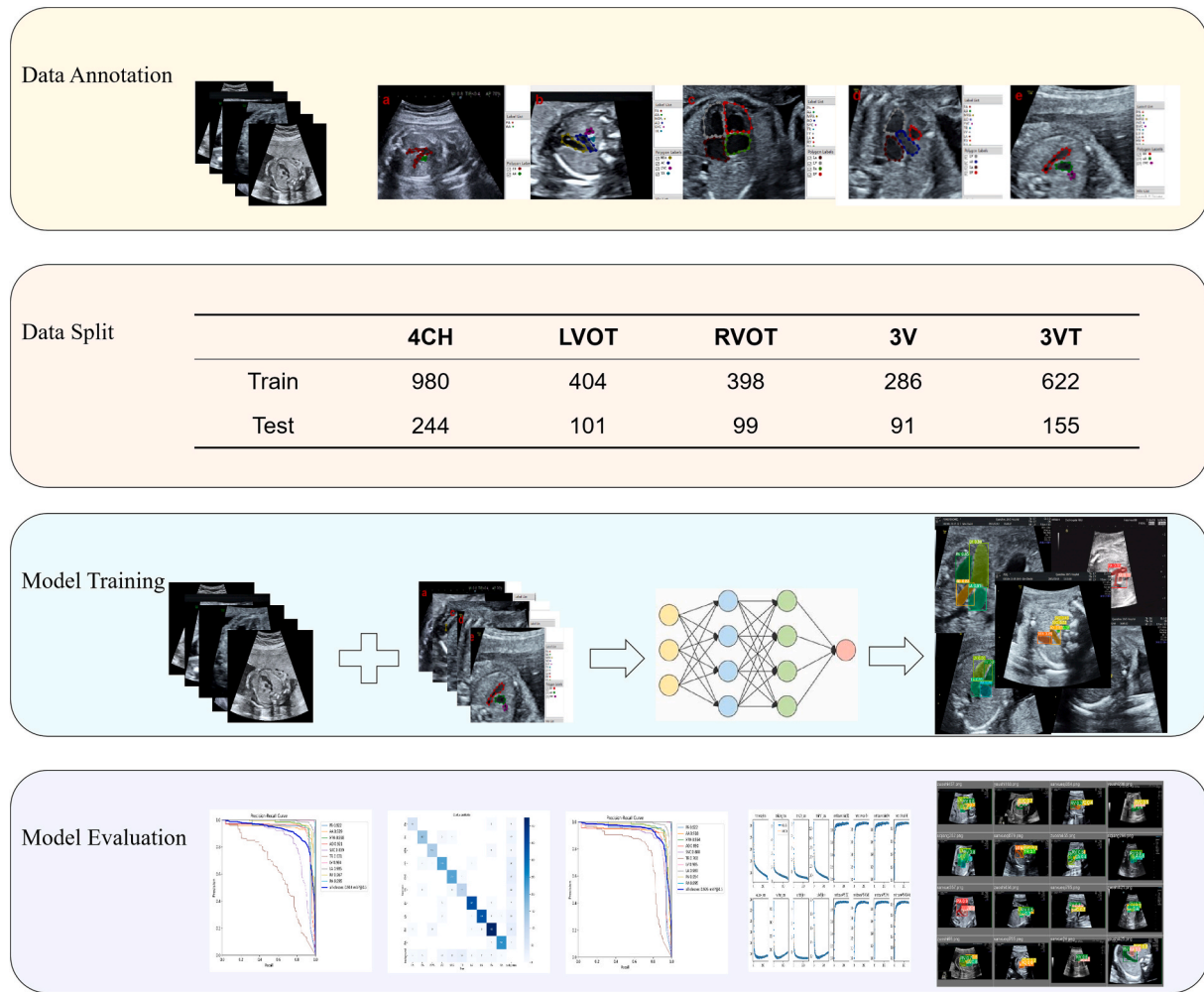


Fig. 2. Overall flow of the experiment.

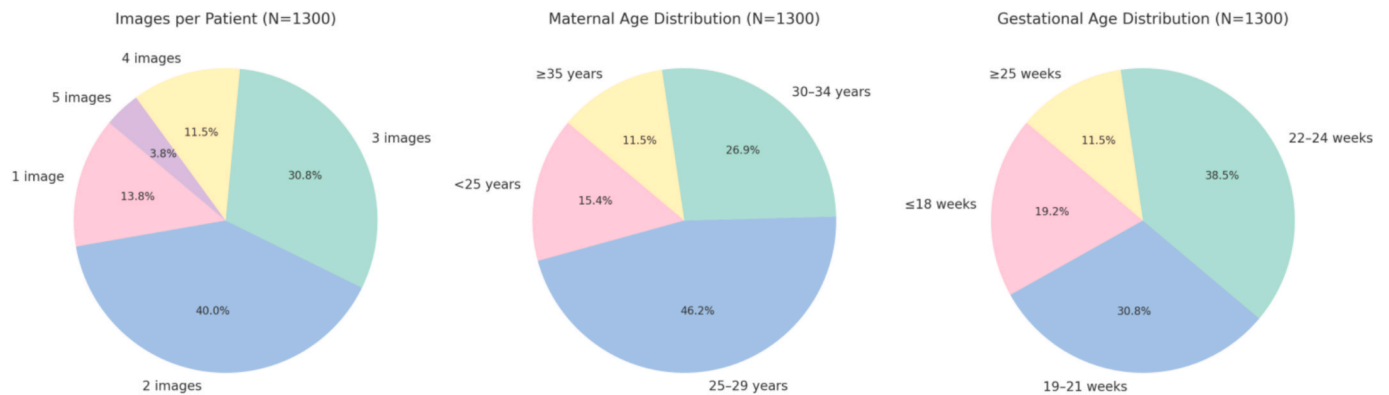


Fig. 3. Statistical distribution of the FH-3 K dataset. (Left) Images per patient. (Middle) Maternal age distribution. (Right) Gestational age distribution.

applicability, and lay a solid foundation for building a more comprehensive and intelligent fetal ultrasound diagnostic system.

3. Materials and methods

Fig. 2 shows the overall flow of this study. First, an expert collected and labeled fetal heart ultrasound images. Second, the organized dataset is subjected to data preprocessing and segmentation. Finally, the fetal heart structure instance segmentation algorithm is trained and

evaluated based on the original images and annotation information in the training and validation datasets.

3.1. FH-3 K dataset

This study's dataset was derived from clinical images obtained from routine heart ultrasonography performed on fetuses in mid-pregnancy (weeks of gestation between $18\text{--}24 \pm 6$) from January 2021 to June 2023. It was recruited from 1,300 pregnant women, with a mean age of

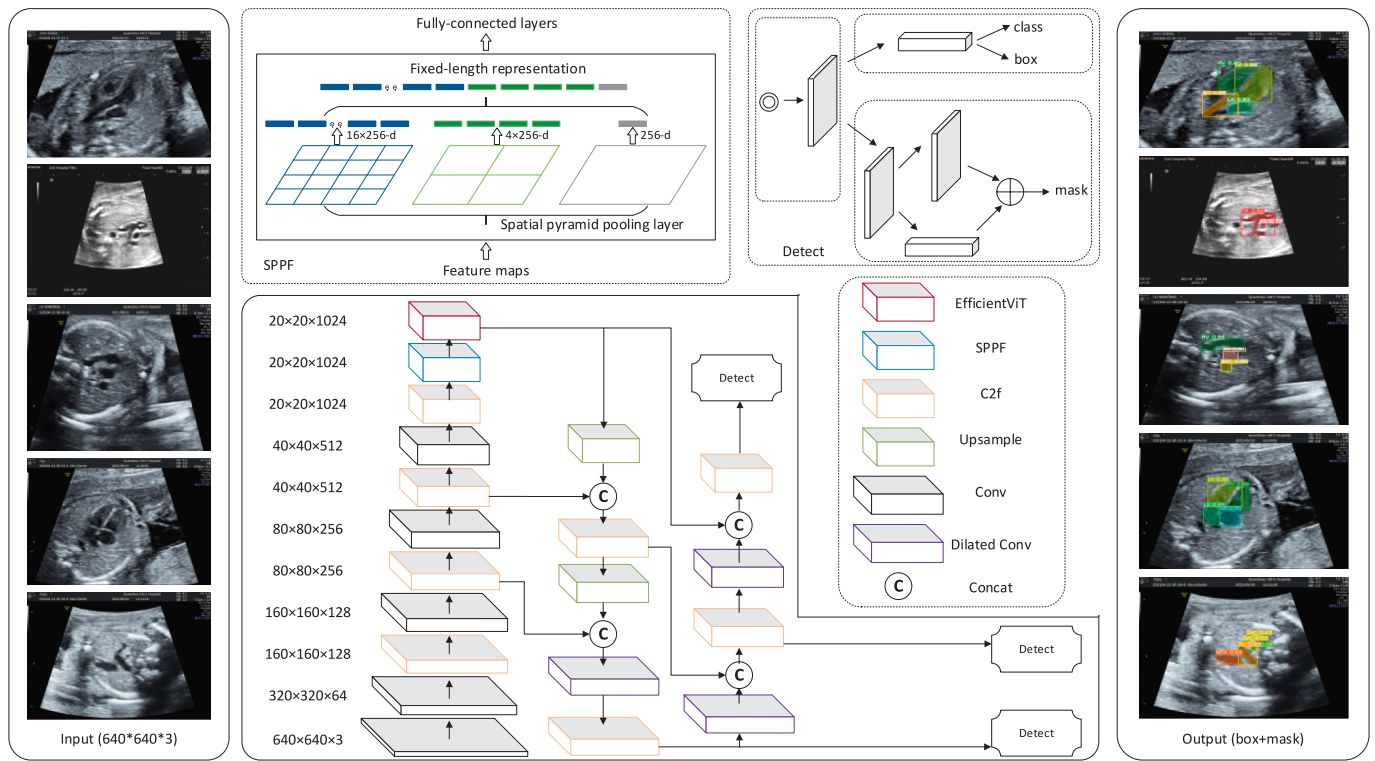


Fig. 4. Overall architecture of the FHSS-NET model.

approximately 29 years, from the People's Hospital Affiliated Of Quanzhou Medical College, the First Hospital of Quanzhou City, and the Second Affiliated Hospital of Fujian Medical University. The study was approved by the Medical Ethics Committee of the Second Affiliated Hospital of Fujian Medical University (Approval No. [2024] Fujian Medical Second Ethics No. 454), and informed consent was obtained from all participants. The project is supported by comprehensive datasets and has undergone thorough clinical validation.

Further statistical results are shown in Fig. 3: (Left) the distribution of images per patient, where most women contributed 2–3 standard ultrasound images; (Middle) the maternal age distribution, indicating that the majority of pregnant women were between 25–34 years old, with smaller proportions younger than 25 or older than 35; and (Right) the gestational age distribution, showing that most fetuses were scanned between 19–24 weeks, consistent with the characteristics of routine mid-pregnancy examinations.

Under strict quality control, the research team collected 3,360 2D ultrasound images from five conventional fetal heart views. It split the data into training and validation sets in an 8:2 ratio. Requirements for inclusion in the sample included complete, clear, and centered target structures in the images, occupying at least half of the frame; no colored blood flow, measurement lines, or noise artifacts; and ultrasound gestational age generally matched actual gestational weeks, with heart and other structural malformations ruled out postnatally. The corresponding images were not recorded if the photos were unstable or if apparent abnormalities were detected due to the subject's size. Image pixel sizes were not standardized in the dataset, with typical resolution ranges covering 704×561 , 720×576 , 720×370 , 768×576 , and 1260×910 . Twelve different brands and models of color Doppler ultrasound diagnostic equipment were used for image acquisition, all of which were equipped with 5.0 MHz convex array probes, which can take into account the real-time scanning of fetal heart rate and structure. To ensure the accuracy of segmentation and labeling, all images were outlined by two experienced ultrasonographers using Labelme[36] software and reviewed by another expert. In case of disagreement between the two

labelers, the photos were jointly discussed and relabeled when necessary, resulting in high-quality example segmentation labeling. This dataset contains standard ultrasound images of fetal heart sections with different devices and image resolutions. It is equipped with accurate labeling based on multiple rounds of expert review and is suitable for researching automatic identification and instance segmentation of fetal heart structures.

3.2. Proposed model

This study proposes an efficient deep network framework, FHSS Net, for instance segmentation of fetal heart anatomical structures.

Fig. 4 illustrates the overall architecture of the proposed FHSS-Net, an efficient instance segmentation network designed for fetal heart ultrasound images. The network takes a $640 \times 640 \times 3$ ultrasound image as input and processes it through successive stages of feature extraction, enhancement, and decoding.

In the backbone, we embed a lightweight EfficientViT module to capture long-range dependencies and global contextual information. This is followed by an SPPF (Spatial Pyramid Pooling – Fast) module, which aggregates multi-scale spatial features using fast-stacked pooling, and C2f (Concatenate-to-Fusion) blocks for residual feature fusion and propagation.

The neck incorporates dilated convolution and upsampling layers to further expand the receptive field and refine spatial resolution, facilitating effective multi-scale feature integration. The fused features from different scales are concatenated and passed into multiple parallel “Detect” heads, which serve as lightweight decoders to generate instance-level predictions.

FHSS-Net employs three parallel Detect modules corresponding to different feature map resolutions. Each Detect module predicts bounding boxes, confidence scores, class probabilities, and mask embeddings. The outputs from all Detect modules are then aggregated and processed through a Non-Maximum Suppression (NMS) layer to remove redundant detections and fuse multi-scale predictions, generating the final

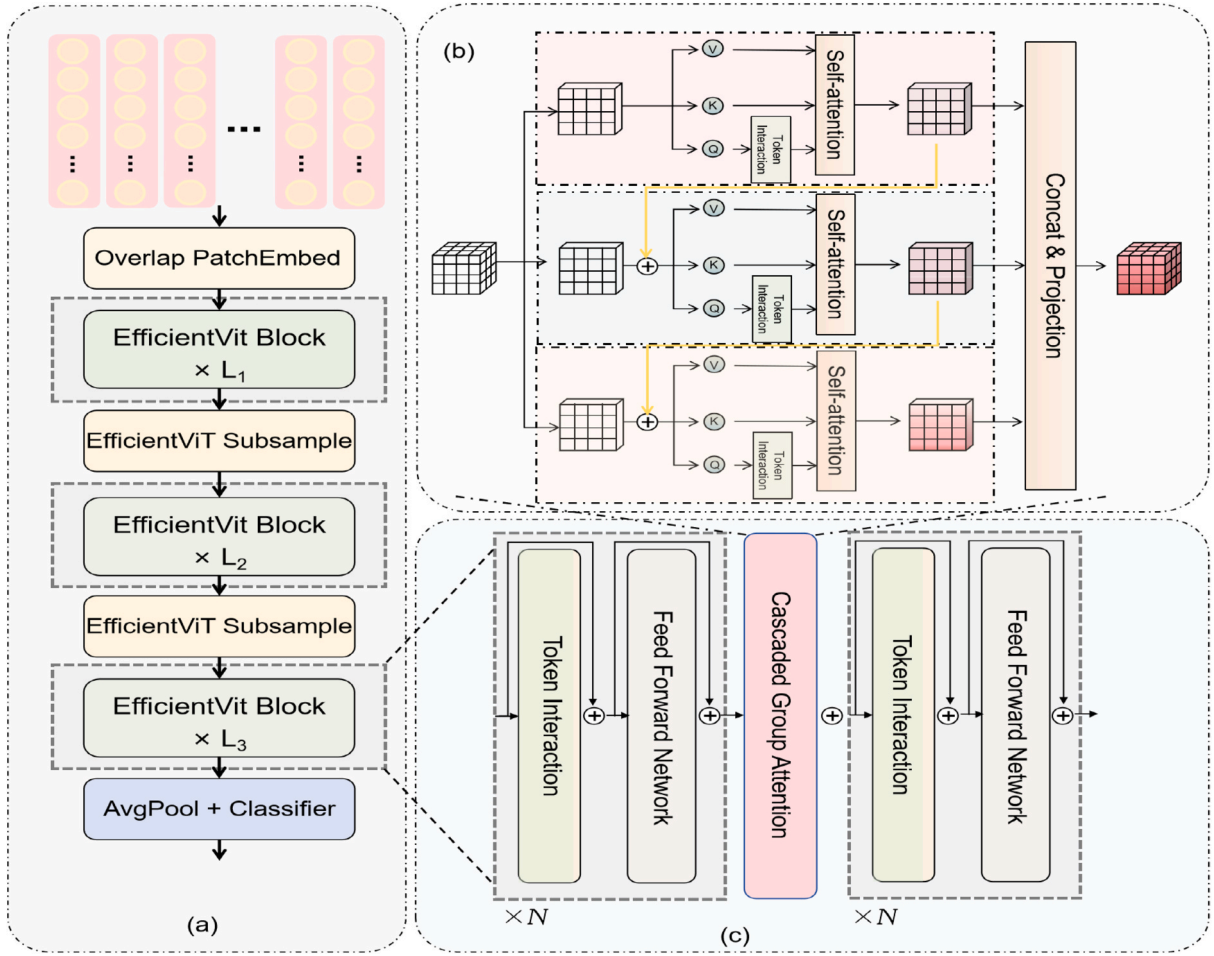


Fig. 5. Detailed view of the EfficientViT module.

instance-level bounding boxes and segmentation masks. Each Detect module outputs bounding boxes and corresponding segmentation masks for the 10 anatomical structures. This multi-head design improves the network's capacity to detect and segment structures of varying sizes and spatial distributions, enabling robust and precise performance in complex ultrasound environments.

As shown in Fig. 4, the model has undergone multiple optimizations to address common issues in ultrasound imaging, such as large-scale differences, substantial noise interference, and blurred edges of details. Firstly, a lightweight EfficientViT module is embedded in the backbone feature extraction stage, which captures long-range dependencies and global contextual information through an efficient self attention mechanism, making the network more sensitive in distinguishing heart chambers, blood vessels, and surrounding tissues; Secondly, dilated convolution was introduced in the feature fusion section to replace traditional convolutional layers, significantly expanding the receptive field without considerably increasing the number of parameters, providing more multi-scale and long-range dependent feature expressions for subsequent segmentation heads, especially helping to enhance the characterization of organ edges and morphologically variable regions; Finally, by combining E-Focal Loss to strengthen the attention to difficult to distinguish samples and small targets, and adjusting the allocation of gradient weights between positive and negative samples, the model can effectively converge and accurately capture key structures in ultrasound environments with imbalanced categories and abundant noise. Overall, FHSS Net has made breakthroughs in the fusion of global and local information, cross-scale feature extraction, and adaptability to complex segmentation scenes,

providing a high-precision and efficient solution for accurately identifying and segmenting 10 anatomical structures of the fetal heart.

3.2.1. EfficientViT module

In FHSS-Net, the EfficientViT module is embedded in the backbone to enhance the network's ability to model long-range dependencies and capture global contextual information in fetal heart ultrasound images. This is particularly important for distinguishing between anatomically adjacent structures such as heart chambers and blood vessels, which often appear with blurred boundaries and low contrast.

EfficientViT is an efficient Vision Transformer (ViT) model that aims to improve traditional Transformer models' computational efficiency and performance when dealing with image tasks. ViT models have received much attention due to their excellent performance on various vision tasks. Still, they usually require a lot of computational resources and data for training. EfficientViT aims to address these issues by introducing key technologies and architectural improvements that make Transformer models more efficient and practical for applications in the vision domain.

The key features and innovations of EfficientViT typically include (1) Model Architecture Optimization: EfficientViT reduces the computational complexity and the number of parameters by improving the model's architecture. This may consist of using more efficient attention mechanisms, such as localized or sparse attention, which reduce the computation required when processing large images. (2) Dynamic Input Resolution: EfficientViT can dynamically adjust the resolution of the input image based on the image's content or the training stage, thereby reducing the amount of computation while maintaining performance.

This approach allows the model to use lower resolution when processing smaller or simpler images and higher resolution when higher accuracy is required. (3) Knowledge Distillation and Transfer Learning: EfficientViT may utilize knowledge distillation techniques to extract useful information from larger pre-trained models and transfer that information to smaller, more efficient models. This helps maintain model performance while reducing its size and computational requirements. (4) Efficient training strategies: to further improve efficiency, EfficientViT may employ efficient training strategies such as progressive sampling or mixed-precision training, which can speed up the training process without sacrificing too much accuracy.

The details of the EfficientViT module are shown in Fig. 5. The overall architecture is shown in Fig. 5(a), which contains (1) Input: the input is an image with height H , width W , and 3 color channels. (2) Stages 1–3: The image passes through three stages of the EfficientViT block. In each stage, the image is further processed and simplified. Each stage contains multiple EfficientViT blocks; L1, L2, and L3 indicate the number of these blocks. The output dimension of each stage is determined by the height, width, and number of channels $C1$, $C2$, and $C3$ of the image, and the spatial resolution (height and width) of the image is reduced in each stage to minimize the amount of computation. (3) Overlapping Block Processing: This step involves features specific to the EfficientViT model, which may include, for example, overlapping patch embeddings and more efficient attention mechanisms. (4) Downsampling: Downsampling the feature maps between different stages can improve computational efficiency. (5) Classification: After passing all EfficientViT blocks, the final classification result is obtained using average pooling (AvgPool) and a classifier.

One of the EfficientViT blocks is shown in Fig. 5(b), and this section details the components and operations inside the EfficientViT block. Each block has two main parts: the Feed Forward Network (FFN) and the Token interaction. These include (1) Token Interaction: this step increases the model's ability to perceive input features. (2) Cascaded Group Attention: This variant of the attention mechanism may be designed to increase processing speed and reduce computational requirements. (3) The operations in each EfficientViT block are repeated N times, which may help to understand the image content at a deeper level.

Fig. 5(c) section shows the cascade group attention module, which details how the cascade group attention mechanism is implemented. This includes (1) Multi-head attention: the input is first segmented into multiple heads, each corresponding to a different representation subspace. In this way, multiple feature information can be acquired simultaneously. (2) Self-attention: A self-attention mechanism in each head allows the model to connect different image parts. (3) Concat & Projection: Finally, the outputs of all heads are concatenated together and projected through a linear layer to produce the output.

In summary, the EfficientViT module in FHSS-Net contributes to improved global feature representation with reduced computational cost. By integrating multi-stage attention and feed-forward interactions, it enhances the network's capability to recognize spatially dispersed and morphologically complex anatomical regions. This leads to more accurate segmentation of the fetal heart structures while maintaining model efficiency.

3.2.2. E-focal loss

E-Focal loss is a novel function based on an improved model of the traditional Focal loss for the category imbalance problem in deep learning. Focal loss is mainly used to solve teaching due to the disparity between the proportion of positive and negative samples in the target detection task difficulty problem [37]. The core idea is to reduce the contribution of easy-to-classify samples to the total loss by adding a moderator so that the model focuses more on difficult-to-identify samples. E-Focal loss is further optimized to achieve a finer balance among different categories. E-Focal loss optimizes the category imbalance in classification tasks by introducing dynamically adjusted balancing

parameters for the classification problem. The method dynamically adjusts the loss weights of each category, taking into account the different sample sizes and classification difficulties of different categories. Its loss function expression is usually defined as follows:

$$L(y, \hat{y}) = -\alpha_t(1 - p_t)^\gamma \log(p_t)$$

Where p_t is the probability that the model predicts the sample to be its proper category, α_t is the weight coefficient for the true category t , and γ is the focusing parameter, which is used to adjust the weight of misclassified samples. The realization process is shown in Algorithm 1.

To enable E-Focal Loss to adapt dynamically to class imbalance and varying classification difficulty, we introduce a systematic strategy for updating both the class-wise balancing factor α and the global focusing parameter γ at each training epoch.

1. Dynamic class-wise Weight α

Let $A_c^{(t)}$ denote the average accuracy of class c in epoch t . Then, the updated class-wise balancing factor $\alpha_c^{(t)}$ is computed as:

$$\alpha_c^{(t)} = \frac{1}{A_c^{(t)} + \epsilon} \bigg/ \sum_{k=1}^C \frac{1}{A_k^{(t)} + \epsilon}$$

where: $A_c^{(t)}$ is the per-class accuracy of class c at epoch t , ϵ is a small constant (e.g., 10^{-6}) added for numerical stability, the normalization ensures $\sum_{c=1}^C \alpha_c^{(t)} = 1$, which stabilizes training.

This mechanism increases the influence of underperforming (i.e., difficult or rare) classes and reduces the dominance of easy-to-classify classes.

2. Dynamic focusing Parameter γ

We further adjust the global focusing parameter γ based on the overall classification difficulty across the dataset at epoch t . Let $E^{(t)}$ denote the misclassification rate:

$$E^{(t)} = 1 - \frac{1}{N} \sum_{i=1}^N 1(y_i = \hat{y}_i)$$

Then the updated focusing parameter is defined as:

$$\gamma^{(t)} = \gamma_{\min} + (\gamma_{\max} - \gamma_{\min}) \cdot E^{(t)}$$

where: $\gamma_{\min}$ and $\gamma_{\max}$ are pre-defined bounds, $\gamma_{\min} = 0.5$, $\gamma_{\max} = 3.0$, N is the total number of training samples, $1(y_i = \hat{y}_i)$ is the indicator function denoting whether the i -th prediction is correct.

This strategy enables E-Focal Loss to emphasize hard samples more strongly when the error rate is high and relax the focus as training progresses and the model improves.

Algorithm 1

Computation of E-Focal Loss.

Input:

y_{true} : Matrix of proper labels, dimension $N \times C$, where N is the number of samples, C is the number of classes

y_{pred} : Matrix of predicted probabilities, dimension $N \times C$

α : Vector of dynamic class weights, dimension C

γ : Dynamic focusing parameter, scalar

Output:

Loss: Computed E-Focal Loss value

Procedure:

1: Initialize total loss to zero, $\text{loss} = 0$

2: For each sample i from 1 to N do:

a. For each class j from 1 to C , do the following:

i. If $y_{\text{true}}[i, j] \neq 1$:

(continued on next page)

Algorithm 1 (continued)

```

- p_t = y_pred[i, j]
- alpha_t = alpha[j]
ii. Else:
- p_t = 1 - y_pred[i, j]
- alpha_t = 1 - alpha[j]
iii. Compute the modulating factor:
- modulating_factor = (1 - p_t)^gamma
iv. Compute the cross-entropy loss for the current class:
- ce = -log(p_t)
v. Compute the contribution of E-Focal Loss for the current sample and class:
- e_focal_contribution = alpha_t * modulating_factor * ce
vi. Accumulate to the total loss:
- loss += e_focal_contribution
3: Compute the average loss over all samples:
- loss = loss / N
Return loss

```

The distinguishing feature of E-Focal loss from traditional Focal loss is its ability to dynamically adjust the sum value according to each category's performance during training. This mechanism helps the model to automatically optimize the loss function when facing different difficulties and several categories and promotes the model in a more balanced direction. Because E-Focal loss can be self-adjusted according to the actual data distribution and model performance, it is highly adaptable to a wide range of different datasets and tasks, especially in the case of extreme category imbalance. By reducing the focus on easy-to-categorize samples and increasing the focus on difficult-to-categorize

samples, E-Focal loss can effectively increase the learning efficiency of the model, accelerate convergence, and improve the model performance on a small number of classes.

3.2.3. Dilated convolution

Due to the fixed receptive field, conventional convolution makes it challenging to capture enough contextual information when processing feature maps with different resolutions, especially for long-range targets or small targets in complex backgrounds, and its feature expression ability is deficient. To solve these problems, FHSS-Net introduces Dilated Convolution in the feature fusion stage, which extends the size of the receptive field by inserting intervals (controlled by the expansion rate r) into the standard convolution kernel while keeping the number of convolution kernel parameters and computational complexity unchanged. The core idea of Dilated Convolution is to sparsely sample the input feature map to capture a more extensive range of contextual information. As shown in Fig. 6(a), traditional convolution samples the input feature map continuously, and its sense field is limited to the physical size of the convolution kernel, which cannot cover a more extensive range. In contrast, as shown in Fig. 6(b), the Dilated Convolution significantly extends the receptive field by inserting $r-1$ dilated pixels between neighboring sampling points, spacing them wider and enabling them to capture more global information. The formula for the Dilate Convolution is:

$$y[i] = \sum_k x[i + r \bullet k] \bullet w[k]$$

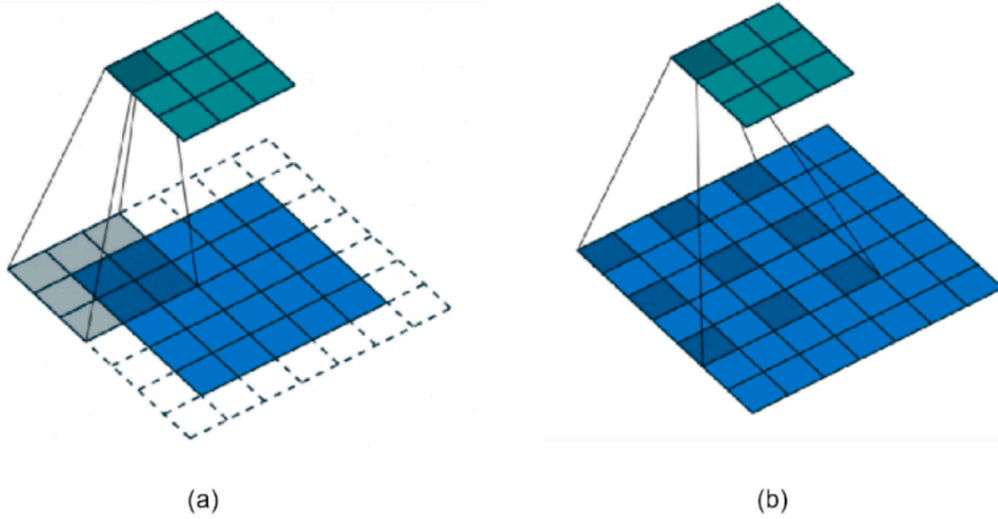


Fig. 6. (a) Traditional convolution (b) Dilated Convolution.

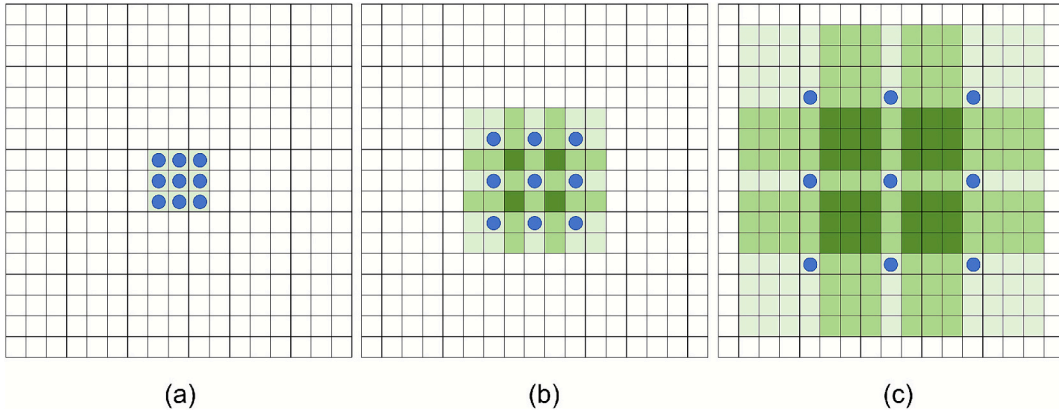
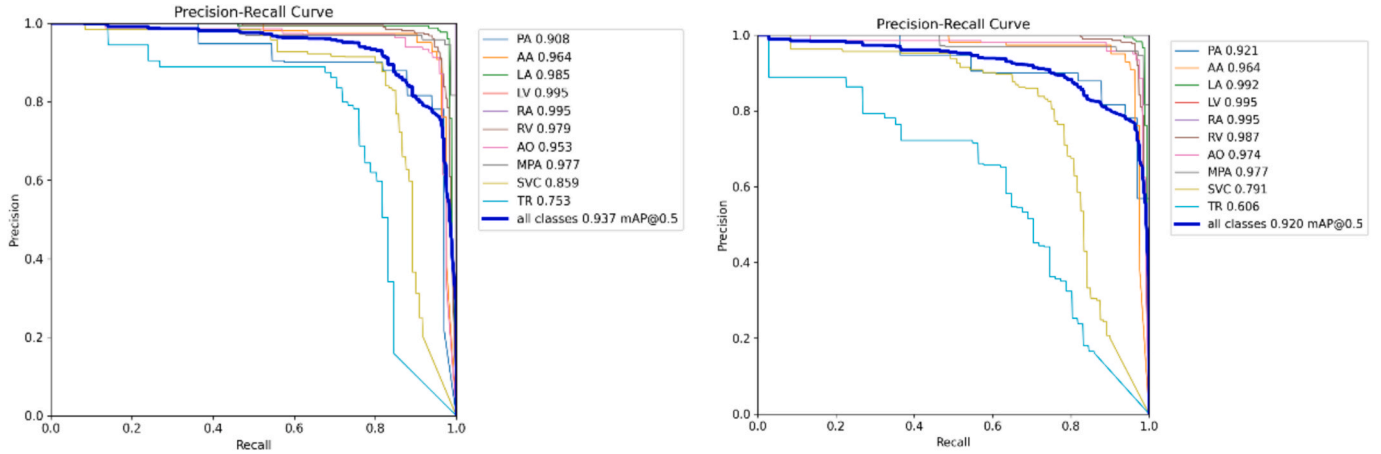


Fig. 7. (a) $r = 1$ (b) $r = 2$ (c) $r = 3$

Table 1

Segmentation accuracy of the FHSS-NET model for key anatomical structures of the fetal heart.

Structures	Box P	R	mAP@0.5	mAP@0.5:0.95	Mask P	R	mAP@0.5	mAP@0.5:0.95	IOU	Dice	NSD	IA
PA	0.805	0.939	0.908	0.592	0.807	0.939	0.921	0.47	0.837	0.911	0.981	0.993
AA	0.917	0.951	0.964	0.58	0.919	0.951	0.964	0.475	0.818	0.900	0.992	0.997
LA	0.958	0.983	0.985	0.676	0.964	0.989	0.992	0.672	0.898	0.947	0.988	0.975
LV	0.996	1	0.995	0.721	0.996	1	0.995	0.686	0.913	0.955	0.984	0.987
RA	0.996	1	0.995	0.763	0.996	1	0.995	0.74	0.925	0.961	0.989	0.996
RV	0.938	0.95	0.979	0.644	0.96	0.964	0.987	0.63	0.887	0.940	0.984	0.980
AO	0.912	0.953	0.953	0.595	0.934	0.975	0.974	0.57	0.851	0.920	0.988	0.974
MPA	0.928	0.986	0.977	0.654	0.929	0.986	0.977	0.633	0.880	0.936	0.988	0.989
SVC	0.893	0.917	0.859	0.374	0.825	0.747	0.791	0.282	0.751	0.858	0.992	0.972
TR	0.859	0.704	0.753	0.265	0.712	0.549	0.606	0.223	0.709	0.830	0.993	0.971
all	0.92	0.928	0.937	0.586	0.904	0.91	0.92	0.538	0.847	0.916	0.988	0.983

**Fig. 8.** P-R curve (Left: P-R curve of Box; Right: P-R curve of Mask).**Table 2**

Comparison of Instance Segmentation Models.

Models	Box_mAP@0.5	Box_mAP@0.5:0.95	Mask_mAP@0.5	Mask_mAP@0.5:0.95	IOU	Dice	NSD	IA	Params (M)	FLOPs (G)	Time (ms/img)
Mask R-CNN	0.769	0.297	0.745	0.284	0.716	0.835	0.901	0.883	44.54	170.2	128.5
PolarMask	0.815	0.368	0.807	0.344	0.739	0.850	0.910	0.911	34.38	110.6	102.8
SOLO	0.85	0.502	0.837	0.403	0.755	0.860	0.928	0.913	20.41	101	88.3
Blendmask	0.873	0.428	0.85	0.417	0.760	0.863	0.932	0.926	37.76	141.1	110.2
YOLOv8	0.890	0.540	0.850	0.495	0.788	0.881	0.957	0.955	3.27	6.06	55.04
YOLACT	0.92	0.531	0.904	0.511	0.791	0.883	0.965	0.955	13.94	57.2	42.5
YOLOv11	<u>0.929</u>	0.554	<u>0.915</u>	0.521	<u>0.817</u>	<u>0.899</u>	0.971	<u>0.977</u>	3.8	6.9	59.8
U-Net++	—	—	—	—	0.779	0.875	0.953	0.949	9.04	34.5	72.3
SwinUNETR	—	—	—	—	0.804	0.891	<u>0.976</u>	0.97	61.48	233.8	186.7
SAM-Med2D	—	—	—	—	0.812	0.896	0.969	0.973	91.1	318.5	232.1
FHSS-NET	0.937	0.586	0.92	0.538	0.847	0.916	0.988	0.983	4	7.8	67.04

Bolding indicates that optimal values were obtained for each metric. Underlined numbers represent the second-best results for each metric.

Where $x[i]$ is the value of the i th pixel point on the input feature, $w[k]$ is the weight of the convolution kernel, r is the expansion rate, and k is the convolution kernel index. When $r = 1$, the null convolution degenerates to standard convolution; and when $r > 1$, $r-1$ null pixels are inserted between neighboring sampling points, and the effective receptive field becomes: $k + (k-1) \cdot (r-1)$. Where k is the original size of the convolution kernel.

For example, for a convolution kernel of 3×3 , when $r = 1, 2, 3$, the practical perceptual fields are 3×3 , 5×5 , 7×7 , respectively, as is shown in Fig. 7. This indicates that as the expansion rate increases, the null convolution can capture a more extensive range of contextual information without additional parameters or computational cost.

In the feature fusion stage of FHSS-Net, different expansion rates are applied to the Dilated Convolution in the fusion path of different resolution feature maps to adapt to different feature requirements. For example, for deep feature maps (low resolution), which are small in resolution and rich in semantic information but lack spatial details, a Dilated Convolution with an expansion rate of $r = 3$ is applied. The effective receptive field is enlarged to 7×7 , which can capture the global contextual information and help in significant target detection or contextual modeling of long-distance targets. The formula is embodied as:

$$y_{low}[i] = \sum_k x_{low}[i + 3 \bullet k] \bullet w[k]$$

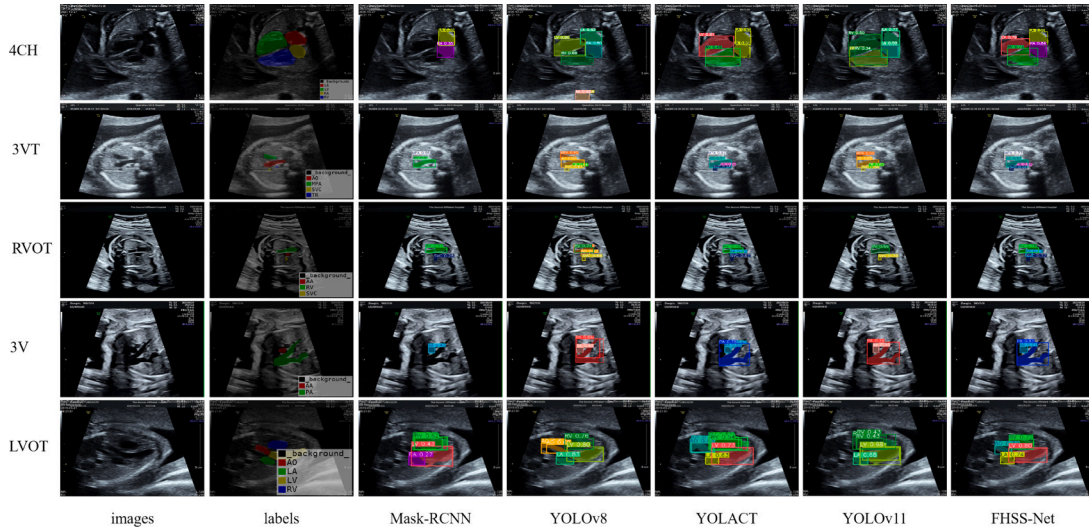


Fig. 9. Comparison of FHSS-NET segmentation results.

Here, $r = 3$ enlarges the receptive field of the deep feature map, allowing it to focus on the global structure of the target.

For medium-resolution feature maps (intermediate layer), Dilated Convolution using the expansion $r = 2$ can take contextual information and local details to some extent, which is especially effective for medium-sized target detection. The formula is:

$$y_{mid}[i] = \sum_k x_{mid}[i + 2 \cdot k] \cdot w[k].$$

Expansion rate $r = 2$ enables the effective receptive field to be extended to 5×5 , which captures semantic information while preserving some of the detailed features.

Spatial details are essential for shallow feature maps (high resolution), mainly to detect small targets. So, standard or null convolution with expansion rate $r = 1$ ensures that sparse sampling does not corrupt fine-grained feature information. Its calculation formula is:

$$y_{high}[i] = \sum_k x_{high}[i + k] \cdot w[k]$$

$r = 1$ at this point, which maintains the integrity and accuracy of the high-resolution feature map. Null convolution at different expansion rates can generate feature maps with multi-scale receptive fields by applying them in the fusion path. Finally, after Concat and up-down sampling operations, these features are integrated into a unified feature pyramid, providing richer feature expressiveness for target classification and bounding box regression.

3.3. Training strategy

A series of data enhancement operations were done before feeding the data into the model, mainly as follows: first, random rotation and flipping techniques were applied to simulate different imaging angles and orientations to increase the diversity of the data. Second, scale transformation and cropping were implemented to accommodate the model's learning of various sizes and details by changing the image size and focusing on different image regions. Again, random noise and blurring effects are introduced to enhance the robustness of the model to changes in image quality. Finally, different imaging conditions are simulated by adjusting the brightness, contrast, and saturation. Combining these data enhancement techniques significantly increases the amount and diversity of training data and effectively improves the model's ability to process ultrasound images and diagnostic accuracy.

To ensure fair and robust evaluation, we adopted a five-fold cross-

validation strategy across all experiments. The entire FH-3 K dataset was randomly divided into five mutually exclusive folds at the patient level, ensuring that images from the same subject were not shared across training and validation sets. In each fold, four parts were used for training and one for validation, and final results were reported as the average across the five folds.

We also analyzed the distribution of standard fetal heart views across the training and test sets. Specifically, the training set included 980 images for the 4CH view, 404 for LVOT, 398 for RVOT, 286 for 3 V, and 622 for 3VT. Correspondingly, the test set contained 244 images for 4CH, 101 for LVOT, 99 for RVOT, 91 for 3 V, and 155 for 3VT. This distribution ensures that the model is exposed to a balanced representation of all five standard views during both training and evaluation.

The experiments were conducted on a computer running Windows 10 with the following hardware configuration: a 12th Gen Intel(R) Core (TM) i5-12400F as the CPU, 16 GB of RAM, and an NVIDIA GeForce RTX3050 with 12 GB of GPU memory. The programming environment used was Python 3.8. The deep learning framework used in the experiments is Pytorch 1.8.

3.4. Evaluation metrics

To comprehensively evaluate the performance of the proposed FHSS-NET model on the task of fetal heart structure segmentation, Precision, Recall, mean Average Precision (mAP), Intersection over Union (IoU), Dice Similarity Coefficient (Dice), Normalized Surface Dice (NSD), and Identification Accuracy (IA) are chosen as the primary evaluation metrics in this paper [38].

Precision is the proportion of actual positive samples recognized as positive by the model, reflecting the model's accuracy in recognizing positive samples. Higher precision indicates a lower misdiagnosis rate, which is especially important for clinical applications to avoid unnecessary mental and economic burdens on patients. The formula calculates the precision:

$$Precision = \frac{TruePositives}{TruePositives + FalsePositives}$$

Recall, on the other hand, measures the proportion of positive samples recognized by the model to all actual positive samples, and this index is critical in medical image processing, especially in disease diagnosis, where a higher recall helps to reduce the leakage of diagnosis and ensures that patients can receive timely and accurate treatment. The formula for the recall rate is:

Table 3
Results of the Ablation Experiment.

Modules	EfficientViT	E -Focal Loss	Dilated Conv	Box_mAP@0.5	Box_mAP@0.5:0.95	Mask_mAP@0.5	Mask_mAP@0.5:0.95	IoU	Dice	NSD	IA	Params (M)	FLOPs (G)	Time (ms/img)
×	×	×	×	0.890	0.540	0.850	0.495	0.788	0.881	0.957	0.955	3.27	6.06	55.04
✓	×	×	×	0.915	0.565	0.875	0.510	0.815	0.895	0.968	0.966	3.65	7.25	59.8
×	×	✓	×	0.903	0.553	0.863	0.505	0.804	0.888	0.963	0.961	3.3	6.1	56.2
×	×	×	✓	0.905	0.555	0.858	0.505	0.801	0.886	0.962	0.958	3.35	6.3	57.1
✓	✓	✓	×	0.925	0.570	0.880	0.520	0.828	0.902	0.975	0.973	3.7	7.35	61.8
×	×	✓	✓	0.920	0.568	0.896	0.515	0.832	0.908	0.979	0.976	3.55	6.6	60.1
✓	✓	×	✓	0.930	0.577	0.914	0.525	0.839	0.912	0.982	0.979	3.85	7.5	63.9
✓	✓	✓	✓	0.937	0.586	0.92	0.538	0.847	0.916	0.988	0.983	4	7.8	67.04

$$Recall = \frac{TruePositives}{TruePositives + FalseNegatives}$$

Mean Average Precision (mAP) is an evaluation index that considers multiple categories and detection thresholds. mAP is obtained by calculating the area under the Precision versus Recall curve to get the average precision (AP) and averaging the AP for all categories to get mAP.

$$mAP = \frac{1}{C} \sum_{c=1}^C AP_c$$

This paper further uses two forms of mAP@0.5 and mAP@0.5:0.95 to evaluate the model performance. Where mAP@0.5 refers to the mAP at an intersection and concurrency ratio (IoU) threshold of 0.5, and mAP@0.5:0.95 indicates that the mAP is computed at intervals of 0.05 over a range from an IoU of 0.5 to 0.95 and the average of these mAPs is taken. This method is more rigorous and can more accurately reflect the detailed performance of the model in segmenting complex fetal heart structures.

IoU directly measures the spatial overlap between prediction P and ground truth G:

$$IoU = \frac{|P \cap G|}{|P \cup G|}$$

Dice coefficient emphasizes region overlap and is defined as:

$$Dice = \frac{2|P \cap G|}{|P| + |G|}$$

Dice is particularly useful in medical image segmentation because it is more robust for small and imbalanced structures. Both metrics are reported per anatomical structure and averaged (Mean IoU, Mean Dice).

NSD evaluates boundary agreement within a clinically relevant tolerance τ (e.g., 1–2 mm). A predicted boundary point is counted as correct if it lies within τ of the ground truth boundary:

$$NSD = \frac{TP_{\tau}}{TP_{\tau} + FP_{\tau} + FN_{\tau}}$$

where TP_{τ} , FP_{τ} , FN_{τ} denote surface points classified within or outside the tolerance margin. This metric reflects whether the segmentation boundary is sufficiently accurate for clinical use.

IA measures whether each anatomical structure is correctly identified. A structure is considered correctly detected if its IoU exceeds a predefined threshold (e.g., 0.5):

$$IA = \frac{\text{Number of correctly identified structures}}{\text{Total number of structures}}$$

This metric reflects clinical usability, as physicians often require reliable identification of all relevant structures rather than perfect pixel alignment.

By combining object-level metrics (Precision, Recall, mAP) with overlap-based metrics (IoU, Dice), boundary-level evaluation (NSD), and identification-level assessment (IA), this study provides a comprehensive and clinically meaningful evaluation framework for fetal heart structure segmentation.

4. Experimental results and analysis

4.1. Experimental results of fetal heart structure segmentation

The final segmentation performance of FHSS-NET on all ten key anatomical structures in fetal heart ultrasound images is shown in Table 1. The overall segmentation accuracy of FHSS-NET reaches 0.937. The best segmentation effect is the left ventricle and the right atrium, with the segmentation and detection of mAP@0.5 reaching 0.995, the

Table 4

Results of the Ablation Experiment.

Loss Function	Box_mAP@0.5	Mask_mAP@0.5	IoU	Dice	NSD	IA
Cross-Entropy	0.875	0.832	0.777	0.875	0.945	0.937
C-IoU Loss	0.890	0.850	0.788	0.881	0.957	0.955
G-IoU Loss	0.901	0.874	0.805	0.892	0.966	0.969
D-IoU Loss	0.905	0.886	0.813	0.897	0.971	0.969
Focal Loss	0.907	0.897	0.829	0.907	0.979	0.975
E-Focal Loss	0.937	0.92	0.847	0.916	0.988	0.983

Table 5

Effect of different numbers of EfficientViT layers.

Layers	Box_mAP@0.5	Mask_mAP@0.5	Params (M)	FLOPs (G)	Time (ms/img)
# 1	0.915	0.88	3.5	7.1	63.8
# 2	0.927	0.898	3.7	7.3	65.5
# 3	0.937	0.92	4	7.8	67.04
# 4	0.935	0.917	4.3	8.1	68.9

Precision value of 0.996, and the Recall value of 1. The worst segmentation effect is the tracheal structure, mAP@0.5, mAP@0.5. From the results analysis, all anatomical structures except the tracheal structure are segmented. The worst impact is the tracheal structure, with only 0.753 for detection mAP@0.5 and 0.606 for segmentation mAP@0.5. The analysis of the results shows that the segmentation effect of all the anatomical structures except the tracheal structure is above 85 %. To observe the segmentation performance of FHSS-NET on different anatomical structures more intuitively, the detection and segmentation P-R curves of FHSS-NET for different anatomical structures were plotted in the test set, as shown in Fig. 8.

4.2. Model comparison results

This study compares with some classical instance segmentation models such as Mask R-CNN[39], SOLO[40], PolarMask[41], and Blendmask[42]. We also include recent representative architectures such as YOLACT[43], YOLOv11, and three popular semantic segmentation models—U-Net++ [44], SwinUNETR [45], and SAM-Med2D [46]—to provide a comprehensive comparison across segmentation paradigms. All models are trained and tested under the same settings: the batch size is set to 8, and training is conducted for 250 epochs using the Adam optimizer with a learning rate of 0.0001 to ensure convergence.

Since SwinUNETR, SAM-Med2D, and U-Net++ are semantic segmentation models, they do not produce instance-level outputs (e.g., bounding boxes or individual masks). Therefore, Box mAP@0.5 and Mask mAP@0.5 are not applicable to these models. For fair evaluation, we compare them using region-based metrics such as IoU, Dice, NSD, and IA.

The results are shown in Table 2, where FHSS-NET achieves the best performance across all available metrics, demonstrating its superior accuracy and efficiency in segmenting fetal heart anatomical structures.

Fig. 9 visualizes the results of several models that perform well for detecting 2D fetal heart ultrasound standard section images. The segmentation results show that for the same fetal heart ultrasound image, all the other models produce some erroneous segmentation; in

comparison, FHSS-NET can better and more accurately segment and detect the key anatomical structures of each fetal heart ultrasound standard section with the lowest error rate.

4.3. Ablation Experiment

Ablation Study on Modules. To comprehensively verify the effectiveness of each proposed module in FHSS-NET, we conduct a series of ablation experiments based on the YOLOv8 [47] instance segmentation framework. The ablations focus on three core improvements: (1) the EfficientViT module for enhancing feature representation; (2) the E-Focal Loss, an improved focal loss with adaptive class-level weighting; and (3) Dilated Convolution, which enlarges the receptive field without sacrificing resolution.

The results are presented in Table 3, which clearly demonstrates that each component contributes to the overall performance improvement. Notably, the combination of all three modules achieves the best results across both accuracy and efficiency metrics.

Ablation Study on Loss Function. To further demonstrate the superiority of E-Focal Loss, we conduct detailed comparative experiments with several commonly used loss functions in detection and segmentation tasks, including Cross-Entropy Loss, Focal Loss, and IoU-based losses (C-IoU, G-IoU, D-IoU). Results are summarized in Table 4.

Ablation Study on EfficientViT Block Layers. To determine the optimal architectural depth for the EfficientViT module, we vary the number of inserted EfficientViT blocks from 1 to 4. As shown in Table 5, three layers offer the best trade-off between accuracy and computation. Further stacking leads to diminishing returns and increased overhead.

Ablation Study on Dilation Rates in Feature Fusion. To investigate the effectiveness of applying different dilation rates (r) in the feature fusion stage of FHSS-Net, we conduct a controlled ablation study. Specifically, we vary the dilation rates applied to the shallow, middle, and deep-level feature maps to evaluate their impact on segmentation performance. Table 6 shows the performance under five configurations, including baseline (no dilation), uniform dilation, reversed assignment, and our proposed hierarchical setting ($r = 1/2/3$).

5. Conclusion and Discussion

This study addresses the challenges of noise interference, blurred anatomical structure boundaries, and significant scale differences in the segmentation of fetal heart ultrasound images. A deep learning framework, FHSS Net, was developed to integrate multiple optimizations. In the feature extraction stage, a lightweight self-attention module is introduced to better balance the network's global perception and computational efficiency, achieving accurate identification of fuzzy

Table 6

Performance comparison of different dilation rate configurations in the feature fusion stage.

ID	Shallow(r)	Middle(r)	Deep(r)	Box mAP@0.5	Mask mAP@0.5	IoU	Dice	Params (M)	FLOPs (G)
1 (no dilation)	1	1	1	0.911	0.897	0.803	0.884	3.27	6.06
2 (uniform dilation)	2	2	2	0.917	0.906	0.811	0.891	3.27	6.06
3 (uniform dilation)	3	3	3	0.913	0.903	0.808	0.889	3.27	6.06
4 (Ours)	1	2	3	0.929	0.915	0.817	0.899	3.27	6.06
5 (reversed)	3	2	1	0.914	0.905	0.81	0.89	3.27	6.06

boundaries and noise interference. In the feature integration stage, dilated convolution technology significantly expands the receptive domain, allowing the model to capture richer contextual information and multi-scale details. Through E-Focal Loss, the focus on unevenly distributed and extremely difficult-to-segment samples is effectively adjusted, enhancing the ability to locate and distinguish key anatomical regions. The experimental results show that FHSS Net achieves significantly better accuracy and stability than traditional methods, for instance segmentation tasks of multiple essential structures in the fetal heart, thoroughly verifying the excellent performance of this framework in dealing with the complexity and heterogeneity of ultrasound images. Overall, this study provides more reliable technical support for the automated analysis and accurate diagnosis of fetal heart ultrasound images through structural improvements from multiple perspectives.

Although FHSS-NET achieves excellent results in several key performance indicators, it still shows limitations when dealing with particularly complex heart structures, such as the trachea. These structures pose additional challenges for accurate segmentation due to their morphological complexity and the lack of apparent boundaries in ultrasound images. Therefore, more in-depth feature extraction and boundary recognition techniques can be further explored in future work to compensate for existing models' shortcomings.

In addition, although the preliminary results of this study validate the model on a standard dataset, in actual clinical applications, the model needs to process ultrasound images from different devices and operating conditions, which may affect its performance and generalization ability. Future research could focus on enhancing the stability and adaptability of the model, for example, by implementing a wider range of data enhancement techniques or developing new algorithms to optimize the model's adaptation to different imaging conditions. Also, exploring the application of models in real-time diagnostic environments is an important direction, and there is a need to ensure that models can process flowing image data quickly and accurately to meet clinical needs. In future extensions, integrating privacy-preserving techniques such as forensic-ready watermarking [48] could help protect sensitive medical ultrasound data and improve traceability. Moreover, the adoption of federated semi-supervised learning frameworks [49] may facilitate collaborative training across institutions without compromising patient data privacy. Furthermore, hybrid architectures like HCMM-Net [50], which combine convolutional layers with MLP-based global context modeling, may serve as alternative lightweight backbones to improve both segmentation accuracy and inference efficiency. In addition, recent research has made progress in ultrasound segmentation of other anatomical structures. (Geng, 2024) proposed a force-sensing-based method for vascular segmentation using sequential ultrasound data, incorporating external force cues to enhance spatio-temporal consistency [51]. However, this study mainly targets dynamic vascular structures, whereas our FHSS-Net focuses on static fetal heart standard plane images, emphasizing multi-scale feature modeling and instance-level anatomical segmentation. The two approaches differ in clinical application scenarios and imaging characteristics, highlighting the complementarity and extensibility of deep learning methods across various types of ultrasound image analysis.

Finally, enhancing model interpretability and user interaction will improve clinical acceptance. Developing interpretive tools and visualization interfaces tailored to specific medical needs can help physicians better understand and trust the model's decision-making process, thus advancing its use in medical practice. With these comprehensive improvements, FHSS-NET is expected to become a powerful tool in diagnosing and treating fetal heart disease, increasing the efficiency of early detection and treatment.

Data availability Statement

A subset of the FH-3 K dataset used in this study has been made publicly available to promote transparency and reproducibility.

Specifically, 50 fetal heart ultrasound images and their corresponding annotation files from each of the five standard planes have been uploaded to the following GitHub repository: <https://github.com/1024Strawberries/FH-3K/releases/tag/FH-3K-V1>.

CRedit authorship contribution statement

Furong Li: Writing – original draft, Methodology. **Jun Wang:** Resources, Data curation. **Yating Hong:** Resources, Data curation. **Xiaying Yang:** Validation, Investigation. **JiaXuan Jiang:** Visualization. **Zhong Wang:** Validation. **Shubin Zeng:** Visualization. **Yuling Fan:** Software. **Guorong Lyu:** Writing – review & editing, Project administration. **Ping Li:** Project administration. **Shaozheng He:** Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The authors do not have permission to share data.

References

- [1] W. Wu, J. He, X. Shao, Incidence and mortality trend of congenital heart disease at the global, regional, and national level, 1990–2017, *Medicine* 99 (2020) e20593.
- [2] H.Y. Sun, Prenatal diagnosis of congenital heart defects: echocardiography, *Translational Pediatrics* 10 (2021) 2210.
- [3] S. Xiao, J. Zhang, Y. Zhu, Z. Zhang, H. Cao, M. Xie, L. Zhang, Application and progress of artificial intelligence in fetal ultrasound, *J. Clin. Med.* 12 (2023) 3298.
- [4] M.C. Fiorentino, F.P. Villani, M. Di Cosmo, E. Frontoni, S. Moccia, A review on deep-learning algorithms for fetal ultrasound-image analysis, *Med. Image Anal.* 83 (2023) 102629.
- [5] J.-X. Jiang, Y. Li, Z. Wang, Structure-aware single-source generalization with pixel-level disentanglement for joint optic disc and cup segmentation, *Biomed. Signal Process. Control* 99 (2025) 106801.
- [6] J.-X. Jiang, W. Lei, Y. Wu, H. Wu, F. Li, Y. Xie, X. Zhang, Z. Wang, MFM-DA: Instance-Aware Adaptor and Hierarchical Alignment for Efficient Domain Adaptation in Medical Foundation Models, *arXiv Preprint arXiv:2503.00802* (2025).
- [7] H. Wu, Y. Yang, H. Xu, W. Wang, J. Zhou, L. Zhu, Rainmamba: Enhanced locality learning with state space models for video deraining, in: *Proceedings of the 32nd ACM International Conference on Multimedia*, 2024, pp. 7881–7890.
- [8] R. Yang, Y. Yu, Artificial convolutional neural network in object detection and semantic segmentation for medical imaging analysis, *Front. Oncol.* 11 (2021) 638182.
- [9] E. Elyan, P. Vuttipittayamongkol, P. Johnston, K. Martin, K. McPherson, C. F. Moreno-García, C. Jayne, M.M.K. Sarker, Computer vision and machine learning for medical image analysis: recent advances, challenges, and way forward, *Artificial Intelligence Surgery* 2 (2022) 24–45.
- [10] J. Anquez, E.D. Angelini, G. Grangé, I. Bloch, Automatic segmentation of antenatal 3-D ultrasound images, *IEEE Trans. Biomed. Eng.* 60 (2013) 1388–1400.
- [11] R. Rahmatullah, M.A. Ma'Sum, P. Mursanto, B. Wiweko, et al., Automatic fetal organs segmentation using multilayer super pixel and image moment feature, in: *2014 International Conference on Advanced Computer Science and Information System*, IEEE, 2014, pp. 420–426.
- [12] L. Vargas-Quintero, B. Escalante-Ramírez, L.C. Marín, M.G. Huerta, F.A. Cosío, H. B. Olivares, Left ventricle segmentation in fetal echocardiography using a multi-texture active appearance model based on the steered Hermite transform, *Comput. Methods Programs Biomed.* 137 (2016) 231–245.
- [13] L. Wu, Y. Xin, S. Li, T. Wang, P.-A. Heng, D. Ni, Cascaded fully convolutional networks for automatic prenatal ultrasound image segmentation, in: *2007 IEEE 14th International Symposium on Biomedical Imaging (ISBI 2017)*, IEEE, 2017, pp. 663–666.
- [14] V. Sundaresan, C.P. Bridge, C. Ioannou, J.A. Noble, Automated characterization of the fetal heart in ultrasound images using fully convolutional neural networks, in:

- 2017 IEEE 14th International Symposium on Biomedical Imaging (ISBI 2017), IEEE, 2017, pp. 671–674.
- [15] X. Yang, L. Yu, S. Li, H. Wen, D. Luo, C. Bian, J. Qin, D. Ni, P.-A. Heng, Towards automated semantic segmentation in prenatal volumetric ultrasound, *IEEE Trans. Med. Imaging* 38 (2018) 180–193.
 - [16] S. Meenakshi, M. Suganthi, P. Sureshkumar, Segmentation and boundary detection of fetal kidney images in second and third trimesters using kernel-based fuzzy clustering, *J. Med. Syst.* 43 (2019) 203.
 - [17] L. Xu, M. Liu, J. Zhang, Y. He, Convolutional-neural-network-based approach for segmentation of apical four-chamber view from fetal echocardiography, *IEEE Access* 8 (2020) 80437–80446.
 - [18] D. Qiao, F. Zulkernine, Dilated squeeze-and-excitation U-Net for fetal ultrasound image segmentation, in: 2020 IEEE Conference on Computational Intelligence in Bioinformatics and Computational Biology (CIBCB), IEEE, 2020, pp. 1–7.
 - [19] Z. Sobhaninia, S. Rafiei, A. Emami, N. Karimi, K. Najarian, S. Samavi, S. R. Soroushmehr, Fetal ultrasound image segmentation for measuring biometric parameters using multi-task deep learning, in: 41st Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC), IEEE, 2019, pp. 6545–6548.
 - [20] P. Li, H. Zhao, P. Liu, F. Cao, Automated measurement network for accurate segmentation and parameter modification in fetal head ultrasound images, *Med. Biol. Eng. Comput.* 58 (2020) 2879–2892.
 - [21] I. Oguz, N. Yushkevich, A. Pouch, B.U. Oguz, J. Wang, S. Parameshwaran, J. Gee, P.A. Yushkevich, N. Schwartz, Minimally interactive placenta segmentation from three-dimensional ultrasound images, *J. Med. Imag.* 7 (2020) 014004.
 - [22] J. Xi, J. Chen, Z. Wang, D. Ta, B. Lu, X. Deng, X. Li, Q. Huang, Simultaneous segmentation of fetal hearts and lungs for medical ultrasound images via an efficient multi-scale model integrated with attention mechanism, *Ultrason. Imaging* 43 (2021) 308–319.
 - [23] K.C. Wong, E.S. Sinkovskaya, A.Z. Abuhamad, T. Syeda-Mahmood, Multiview and multiclass image segmentation using deep learning in fetal echocardiography, in: *Medical Imaging 2021: Computer-Aided Diagnosis*, SPIE, 2021, pp. 308–313.
 - [24] S. Budd, M. Sinclair, T. Day, A. Vlontzos, J. Tan, T. Liu, J. Matthew, E. Skelton, J. Simpson, R. Razavi, others, Detecting hypo-plastic left heart syndrome in fetal ultrasound via disease-specific atlas maps, in: *Medical Image Computing and Computer Assisted Intervention—MICCAI 2021: 24th International Conference, Strasbourg, France, September 27–October 1, 2021, Proceedings, Part VII 24*, Springer, 2021, pp. 207–217.
 - [25] R. Liu, M. Liu, B. Sheng, H. Li, P. Li, H. Song, P. Zhang, L. Jiang, D. Shen, NHBS-Net: a feature fusion attention network for ultrasound neonatal hip bone segmentation, *IEEE Trans. Med. Imaging* 40 (2021) 3446–3458.
 - [26] Y. Zeng, P.-H. Tsui, W. Wu, Z. Zhou, S. Wu, Fetal ultrasound image segmentation for automatic head circumference biometry using deeply supervised attention-gated V-Net, *J. Digi. Imag.* 34 (2021) 134–148.
 - [27] V. Singh, P. Sridar, J. Kim, R. Nanan, N. Poornima, S. Priya, G.S. Reddy, S. Chandrasekaran, R. Krishnakumar, Semantic segmentation of cerebellum in 2D fetal ultrasound brain images using convolutional neural networks, *IEEE Access* 9 (2021) 85864–85873.
 - [28] L.S. Hesse, M. Aliasi, F. Moser, M.C. Haak, W. Xie, M. Jenkinson, A.I. Namburete, I.-21st Consortium, others, Subcortical segmentation of the fetal brain in 3D ultrasound using deep learning, *Neuroimage* 254 (2022) 119117.
 - [29] C. Shobana Nageswari, V. Kumar, N. Vini Antony Grace, J. Thiagarajan, Tunicate swarm-based grey wolf algorithm for fetal heart chamber segmentation and classification: a heuristic-based optimal feature selection concept, *J. Intell. Fuzzy Syst.* 44 (2023) 1029–1041.
 - [30] B. Pu, Y. Lu, J. Chen, S. Li, N. Zhu, W. Wei, K. Li, MobileUNet-FPN: a semantic segmentation model for fetal ultrasound four-chamber segmentation in edge computing environments, *IEEE J. Biomed. Health Inform.* 26 (2022) 5540–5550.
 - [31] S. Belciug, Autonomous fetal morphology scan: deep learning+ clustering merger—the second pair of eyes behind the doctor, *BMC Med. Inf. Decis. Making* 24 (2024) 102.
 - [32] Y. Lu, K. Li, B. Pu, Y. Tan, N. Zhu, A YOLOX-based deep instance segmentation neural network for cardiac anatomical structures in fetal ultrasound images, *IEEE/ACM Trans. Comput. Biol. Bioinf.* (2022).
 - [33] S. Sengupta, S. Chakrabarty, R. Soni, Is SAM 2 Better than SAM in Medical Image Segmentation?, *arXiv Preprint arXiv:2408.04212* (2024).
 - [34] F. Li, P. Li, X. Wu, P. Zeng, G. Lyu, Y. Fan, P. Liu, H. Song, Z. Liu, FHUSP-NET: a multi-task model for fetal heart ultrasound standard plane recognition and key anatomical structures detection, *Comput. Biol. Med.* 168 (2024) 107741.
 - [35] S. Qiao, S. Pang, Y. Sun, G. Luo, W. Yin, Y. Zhao, S. Pan, Z. Lv, SPReCHD: Four-chamber semantic parsing network for recognizing fetal congenital heart disease in medical metaverse, *IEEE J. Biomed. Health Inform.* 28 (2022) 3672–3682.
 - [36] B.C. Russell, A. Torralba, K.P. Murphy, W.T. Freeman, LabelMe: a database and web-based tool for image annotation, *Int. J. Comput. Vis.* 77 (2008) 157–173.
 - [37] T.-Y. Ross, G. Dollár, Focal loss for dense object detection, in: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 2017, pp. 2980–2988.
 - [38] A. Reinke, M.D. Tizabi, M. Baumgartner, M. Eisenmann, D. Heckmann-Nötzel, A. E. Kavur, T. Radsch, C.H. Sudre, L. Acion, M. Antonelli, others, Understanding metric-related pitfalls in image analysis validation, *Nat. Methods* 21 (2024) 182–194.
 - [39] K. He, G. Gkioxari, P. Dollár, R. Girshick, Mask r-cnn, in: *Proceedings of the IEEE International Conference on Computer Vision*, 2017, pp. 2961–2969.
 - [40] X. Wang, T. Kong, C. Shen, Y. Jiang, L. Li, Solo: Segmenting objects by locations, in: *Computer Vision—ECCV 2020: 16th European Conference, Glasgow, UK, August 23–28, 2020, Proceedings, Part XVIII 16*, Springer, 2020, pp. 649–665.
 - [41] E. Xie, P. Sun, X. Song, W. Wang, X. Liu, D. Liang, C. Shen, P. Luo, PolarMask: Single shot instance segmentation with polar representation, in: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2020, pp. 12193–12202.
 - [42] H. Chen, K. Sun, Z. Tian, C. Shen, Y. Huang, Y. Yan, BlendMask: Top-down meets bottom-up for instance segmentation, in: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2020, pp. 8573–8581.
 - [43] D. Bolya, C. Zhou, F. Xiao, Y.J. Lee, Yolact: Real-time instance segmentation, in: *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2019, pp. 9157–9166.
 - [44] Z. Zhou, M.M.R. Siddiquee, N. Tajbakhsh, J. Liang, Unet++: A Nested U-Net Architecture for Medical Image Segmentation, in: *Deep Learning in Medical Image Analysis and Multimodal Learning for Clinical Decision Support*, Springer, 2018: pp. 3–11.
 - [45] A. Hatamizadeh, V. Nath, Y. Tang, D. Yang, H. Roth, D. Xu, Swin UNETR: Swin Transformers for Semantic Segmentation of Brain Tumors in MRI Images, *arXiv Preprint arXiv:2201.01266* (2022).
 - [46] J. Cheng, J. Ye, Z. Deng, J. Chen, T. Li, H. Wang, Y. Su, Z. Huang, J. Chen, L.J.H. Sun, J. He, S. Zhang, M. Zhu, Y. Qiao, SAM-Med2D, (2023).
 - [47] G. Jocher, A. Chaurasia, J. Qiu, Ultralytics YOLO, (2023). <https://github.com/ultralytics/ultralytics>.
 - [48] S. Qiao, Q. Guo, M. Wang, H. Zhu, J.J. Rodrigues, Z. Lyu, FRW-TRACE: Forensic-Ready Watermarking Framework for Tamper-Resistant Biometric Data and Attack Traceability in Consumer Electronics, *IEEE Trans. Consum. Electron.* (2025).
 - [49] S. Qiao, M. Huang, H. Li, L. Wang, W. Yin, Y. Sun, Z. Zhao, Z. Lyu, FedSSH: a Consumer-Oriented Federated Semi-Supervised Heterogeneous IoMT Framework, *IEEE Trans. Consum. Electron.* (2025).
 - [50] S. Qiao, S. Pang, P. Xie, W. Yin, S. Yu, H. Gui, M. Wang, Z. Lyu, HCMMNet: Hierarchical Conv-MLP-mixed network for medical image segmentation in metaverse for consumer health, *IEEE Trans. Consum. Electron.* 70 (2023) 2078–2089.
 - [51] Y. Geng, G. Meng, M. Chen, et al., Force Sensing Guided Artery-Vein Segmentation via Sequential Ultrasound Images[C]//International Conference on Medical Image Computing and Computer-Assisted Intervention, Springer Nature Switzerland, Cham, 2024, pp. 656–666.